CS3003: IT Project Management

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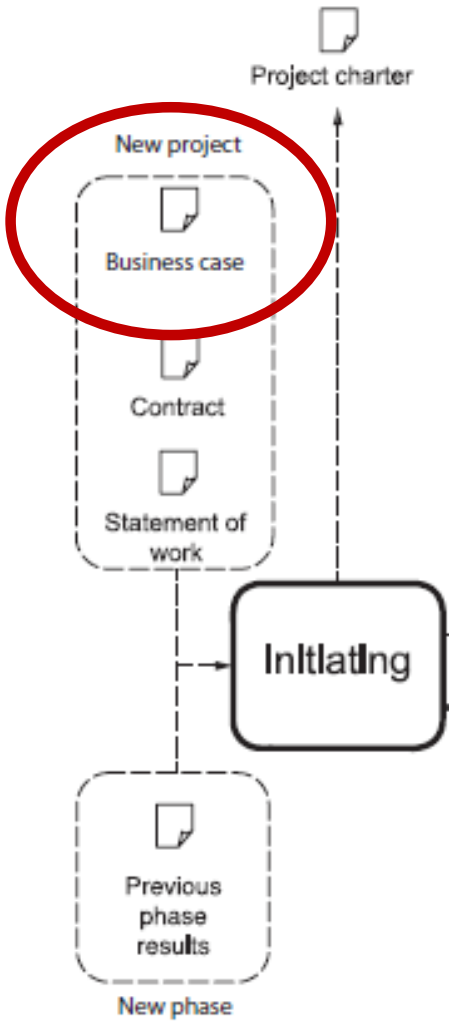
Part 2.1. Project Pre-Initiation:

Business Case, Investment Appraisal, Estimations and Budgeting

Business Case



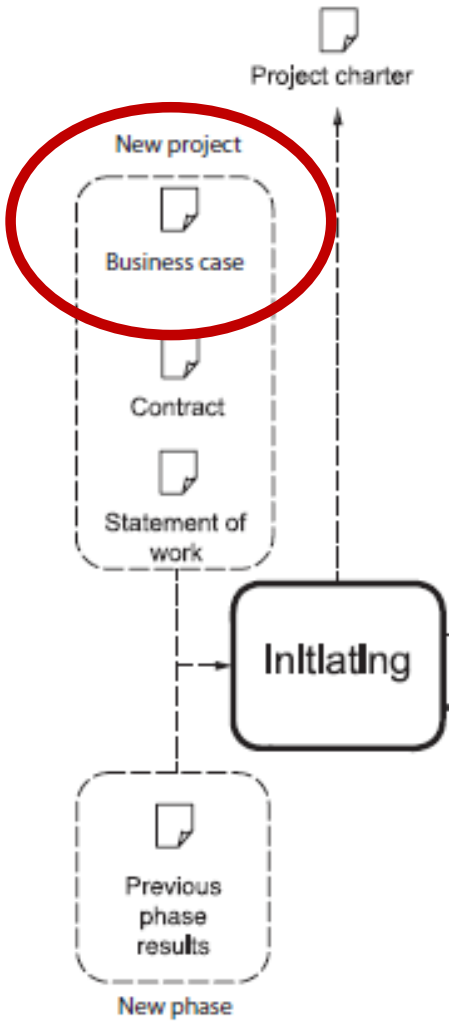
Purpose:

"To document the justification for the undertaking of a project based on the estimated cost of development and implementation against the risks and the anticipated business benefits and savings to be gained. The total business change must be considered, which may be much wider than just the project development cost." (Prince2).

Determine that the estimated effort and time will be worth the expenditure.

The ongoing viability of the project against the Business Case should be monitored during the project.

Business Case



Typical Elements of a Business Case
1. Problem statement and project objectives
2. Summary of scope and any alternatives/options
3. Costs and Investment Appraisal
4. Benefits expected (quantitative, i.e. expressed in measurable terms against today's situation, and qualitative)
5. Risks (summary of the key risks of the project)
6. Timescales (summary of plan)
7. Assumptions/Constraints

Rule of Thumb: Should tell a "compelling story", supported by facts/evidence!

Investment Appraisal

1. **CAPEX** and **OPEX**

Capital expenditures (CAPEX) is funds (money) spent in order to create future benefits for an organization, i.e. purchasing new fixed assets (e.g. equipment, property, buildings) or extending the value (useful life) of existing assets beyond the taxable year.

An **Operating Expense (OPEX)** is any on-going cost for running (operating) the business (e.g. property rent, office stationary, utility bills, travel expenses, personnel salaries).

For example, the purchase of a printer is a CAPEX cost whereas the annual expenses for paper, toner, power and the maintenance of the printer are OPEX costs. Other common OPEX examples are the cost of personnel and facility expenses (e.g. rent, electricity). For accounting/tax purposes, *capital expenditures* must be ***amortized/depreciated*** over the span of several years. Example: the straight-line depreciation of a computer over 4 years, if it was purchased at a cost of £1000 and will have a salvage value of £400, is £150 per year ((1000 — 400)/ 4 years)

Book Value (Beginning of Year)	Depreciation Expense	Accumulated Depreciation	Book Value (End of Year)
1000 (original cost)	150	150	850
850	150	300	700
700	150	450	550
550	150	600	400 (scrap/salvage value)

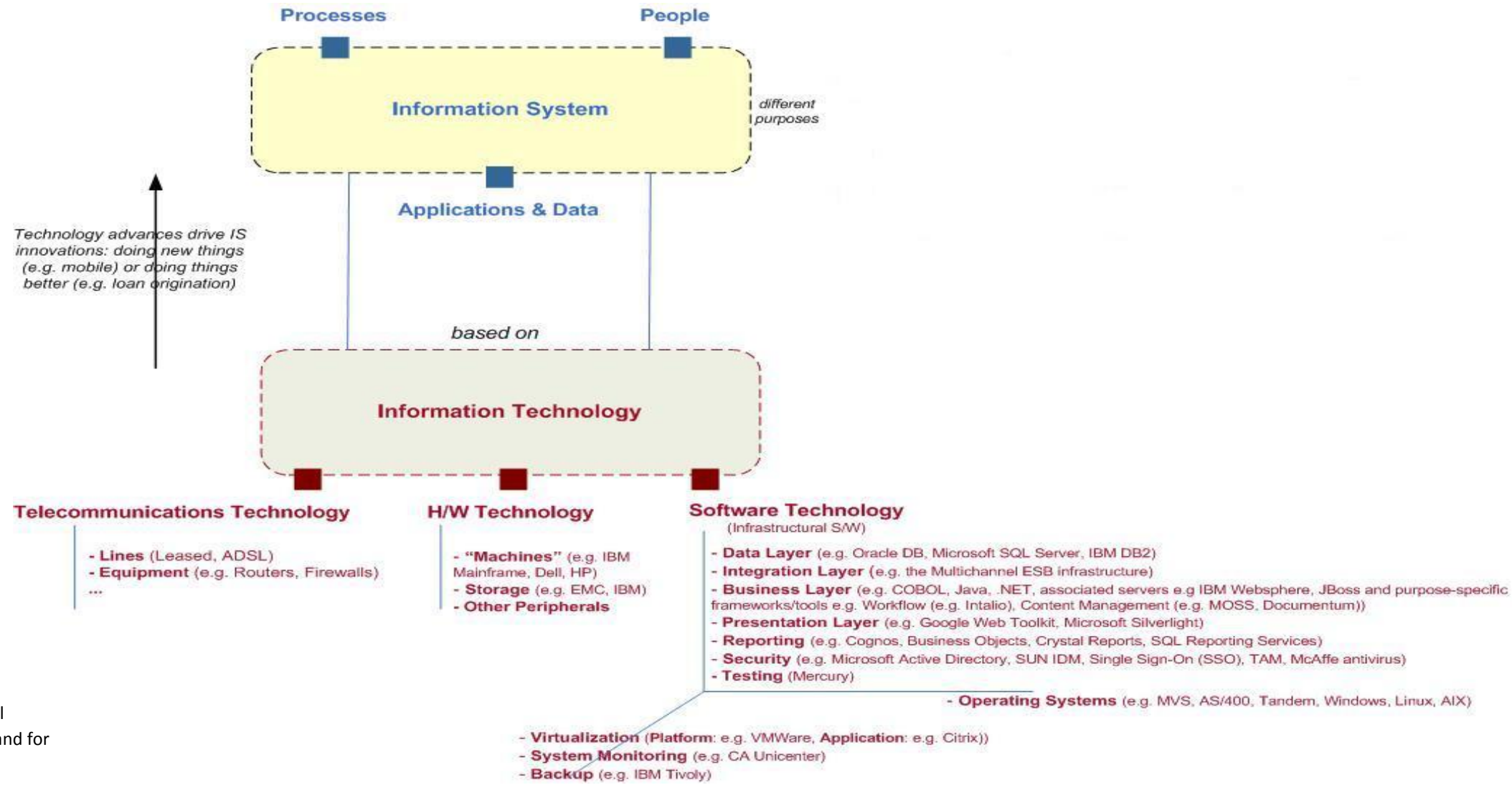
Investment Appraisal

2. Total Cost of Ownership (TCO)

In business decision-making it is important to be able to determine the costs (direct and indirect) of assets and/or to have a cost basis for establishing the total (financial) value of an investment.

A common means used for that purpose is called the *Total Cost of Ownership (TCO),* i.e. a financial estimate whose puprose is to determine the *cost* of *acquiring* and *maintaining* (*operating*) an asset (e.g. a product or IT system) over a number of years.

Investment Appraisal: Total Cost of Ownership (TCO) in IT Projects



Note: The names of commercial products herein are indicative and for educational purposes only.

Investment Appraisal: Total Cost of Ownership (TCO) in IT Projects

Main **Acquisition Cost (CAPEX)** elements:

1. **S/W licenses**
   i. Application SW licenses
   ii. Software Technology related licenses
2. **H/W Technology** (e.g. servers, storage, workstations, appliances)
3. **Telecommunications Technology and Lines**
4. **Acquisition research/decision/procurement costs**
5. **Implementation/Installation Services**
6. **Development and Customization Services**
7. **Integration Services**
8. **Testing**
9. **Consulting**
10. **Training**
11. **Travel**

Main **Operation Cost (OPEX)** elements:

1. **S/W maintenance**
   i. Application SW
   ii. Software Technology related
   iii. Development/Customizations maintenance
2. **H/W Technology maintenance**
3. **Telecommunications Technology and Lines**
4. **Recurring Data Center Costs**, e.g. floorspace, electricity
5. **Recurring Operations Costs** e.g. backups, patches.
6. **Training**

Note 1: Some costs should be considered per target environment, i.e. Live/Production, Disaster, Testing, Development

Note 2: Internal FTE's effort/cost as CAPEX or OPEX. Depreciation?

Investment Appraisal

3. Return on Investment (ROI)

When applied to projects, ROI is a measure (a calculation/estimate) of the value of a project – the benefits received from the project against the total costs of the project over a timeframe so that to know the value of an investment before spending time and money on it.

ROI is usually expressed as a number (percentage) on an annual basis. For example, in its simple form, £100,000 is invested and it earns £120,000. This gives a ROI of 20%.

Note that this is an average number and if the investment spans several years it ignores the “time value of money” i.e. the fact that the current value of an amount of money differs to the value of the same amount of money in the future.

Investment Appraisal

4. Payback Period (PP)

The length of time taken to repay the initial capital cost i.e. “in what timeframe does a project pay for itself”?

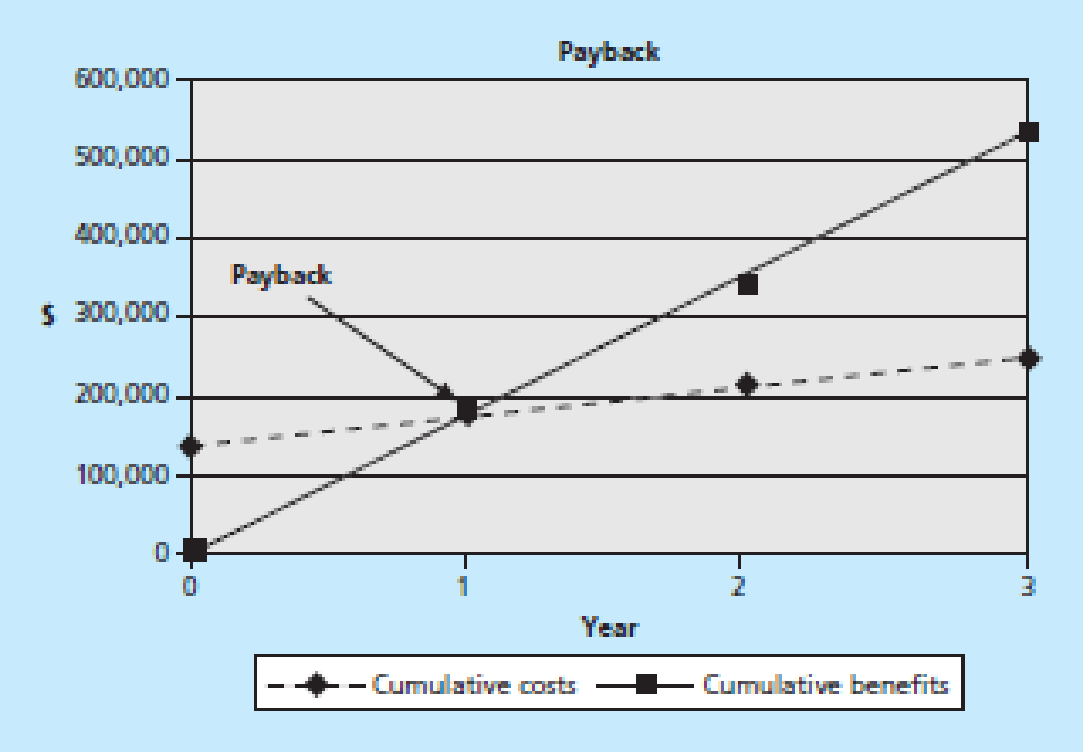


Image from Schwalbe’s book .

Note that the PP method does not take into account the time value of money.

Investment Appraisal

5. Net Present Value (NPV) and Discounted Cash Flows Valuation

"How much does this investement worth in today's money?" — takes into account the time value of money.

Allows evaluating an investment by valuing at the same point in time, i.e. the present, the cash payments and cash receipts expected to be made over the lifetime of the investment.

- **Future Value**: the value at a future date of an amount invested today.

  Future Value = Amount_Invested * $(1+\text{Interest_Rate})^N$, where N is the number of periods we are measuring in the future

- **Present Value:** the value today of a payment that is promised in the future (or else how much should be invested today in order to realize the future value)

  Present Value = Future_Amount / $(1+\text{Interest_Rate})^N$

The higher the interest rate, the lower the present value. Therefore, the interest rate in the present value calculation is called the *discount rate*. The further in the future, the lower the present value.

Investment Appraisal

5. Net Present Value (NPV) and Discounted Cash Flows Valuation (cont.)

- Net Present Value (NPV): is the sum of the present values (PV) of all Net Cash Flows during the period into consideration for the investment
- Net Cash Flows are the Inflows minus Outflows where: inflows are the funds received and outflows are the funds paid out.

- NPV is the sum of all terms $A_t / (1+r)^t$, where,
  - t: the time of the cash flow,
  - r: the discount rate
  - A_t :the net cash flow (inflow minus outflow) at time t

$$\mathrm{NPV} = \sum_{t=0\ldots n} \mathrm{A_t}/(1+r)^t$$

Investment Appraisal

5. Net Present Value (NPV) and Discounted Cash Flows Valuation (cont.).

Example with £100,000 initial investment, 15% discount rate and the net cash flows as in the 2nd column of the table below.

Year	Net Cash Flow (NCF)	Discount Factor (DF): $1/(1+r)^t$ where r=0,15	Present Value (PV) = (NCF x DF)
t=0	-100 000	$1/(1+0,15)^0 = 1$	-100 000
t=1	195 000	$1/(1+0,15)^1 = 0.869$	169 565
t=2	195 000	$1/(1+0,15)^2 = 0.7561$	147 448
t=3	170 000	$1/(1+0,15)^3 = 0.6575$	111 778
t=4	195 000	$1/(1+0,15)^4 = 0.5717$	111 492
t=5	195 000	$1/(1+0,15)^5 = 0.4971$	96 949
t=6	170 000	$1/(1+0,15)^6 = 0.4323$	73 496
t=7	195 000	$1/(1+0,15)^7 = 0.3759$	73 308
t=8	195 000	$1/(1+0,15)^8 = 0.3269$	63 746
t=9	170 000	$1/(1+0,15)^9 = 0.2842$	48 325
t=10	195 000	$1/(1+0,15)^{10} = 0.2471$	48 201
Total	1 775 000		NPV= sum of PV numbers above = 844 308

Project Cost Estimation and Budgeting

Types of Cost Estimates

Estimating Level	Precision Range	When to use this Estimating Level
Order of Magnitude (ROM)	+75, -25 %	When only very basic information is available or needed. Typically used before a project is officially started, e.g. to help managers screen concepts or to make project selection decisions
Budgetary	+25, -10 %	Allocate money into an organization's budget, typically 1-2 years before the project starts. Scope, high-level requirements and resources information is relatively available.
Definitive	+10, -5 %	When detailed information is available, e.g. to make purchasing decisions, to update budget, etc. Typically <1 year before the project starts.

Project Estimation and Budgeting

Main Estimation Approaches

Type	Summary	Advantages	Disadvantages/Issues
Analogous (Top-Down)	Use the actual cost of a previous similar project as a basis for estimating the costs of the current project	+ Simple, quick and low effort (i.e. for ROM estimates). +Can be used before knowing system design details.	Requires similar project and evaluation of the degree of similarity (both functional and technological).
Bottom-Up	Estimating the costs of individual tasks/activities and summing them to get the total. Size of tasks/activities and experience of the estimators drive the accuracy levels.	+Level of Accuracy	Requires more detailed analysis (e.g. tasks/WBS) and time to complete.
Parametric	Use project characteristics (parameters) in a mathematical model (e.g. Function Point Analysis, COCOMO II, or custom estimators)	+Simple +Sometimes a low cost approach	Requires a historic model Depends on quality of model data Depends on quality of input data/parameters

Project Cost Estimation and Budgeting

Additional Approaches

- **Delphi Method or Wideband Delphi:** seek **Expert Advice,** usually from a number of experts until they agree on a common estimate.
- **Three-point Estimate:** estimate the Most Probable (MP), Optimistic (O) and Pessimistic (P) cost for an item. Next use a weighted formula, e.g. the PERT formula,

$$E = (4*MP + O + P) / 6$$

Project Cost Estimation and Budgeting: Important Rules of Thumb/Checklist

1. In your estimation deliverable, explicitly state the underlying **Scope/Details** and any applicable **Assumptions** and **Rationale**.
2. **Be honest!**
3. **Do not do it alone**, i.e. do not tell people how long it will take them to do their work! Gain team and subcontractors consensus before you deliver your estimate.
4. **Use historical information**, but adjust accordingly. Hence, **keep quality cost records of past projects**!
5. Use a **combination** of **both methods** and **people** to estimate.
6. Distinguish **Effort Vs Duration**. In the latter, do not forget holidays!
7. Be aware of the **organizational rules on effort/costs** (e.g. taxes (VAT), labour unit costs, mdays/hours, working days)
8. Be aware of the **organizational rules on controls/required deliverables**, e.g. security checks, legal checks, preparation of manuals, training. These will imply additional tasks/effort.
9. Adjust estimates for **loading factors** (e.g. multiple groups involved, remote locations, travelling, culture/language barriers)
10. Include **reasonable Reserves** (for unanticipated/changing circumstances). Unreasonable reserves have impact (why?)
11. Do not forget effort to: (i) gather detailed, agreed requirements, (ii) test (all types of testing), (iii) setup system in different environments (and have test data available), (iv) number and complexity of interfaces to other systems.
12. **Human beings are biased towards underestimation**. Do not be too optimistic based on your own abilities.
13. If **imposed with infeasible deadlines and budget from management, do not merely accept! Negotiate focusing on the equilibrium of scope/time/quality/cost (rather than cost alone)**! Moreover, be cautious on initial estimates as management never forget these, but rarely remember subsequent changes/adjustments. Keep them informed formally and keep records. **Be prepared to be challenged on your estimates, hence make sure they are defensible**!

Project Cost Estimation and Budgeting

Project Budget: The process and relevant artifact of allocating the estimated costs, labor and material, over time. The main goal is to establish an *authorized cost baseline* and the key benefit is that it determines the cost baseline against which project performance can be monitored and controlled. It also facilitates the organization's overall costs and cash flow management.

- Detailed Budget preparation is typically part of the Project Planning process rather than the Pre-Initiation phase. However, in practice, it is common that a project allocated budget exists at this stage (e.g. with a high-level allocation in time)
- Apart from the cost estimates, the budgeting process has a number of inputs which will be discussed later (e.g. WBS, project plan/schedule, actual project team, risks).
- It also typically includes Reserves (Contingency and Management Reserves)
- Most organizations have their own process and rules on how to prepare project budgets (e.g. on the way to include FTE costs, on depreciation, on taxation, on currency).
- Budgets have *deviations*, which should be managed (as part of what is known as Project Cost Management), and therefore could change over the course of the project – such changes should be formally approved and communicated.

CS3003: IT Project Management

Part 2.2 Project Pre-Initiation:

IT Systems Acquisition Models and Procurement Management

IT Systems Main Acquisition Models

Custom-built
- In-house
- Outsourced /Offshored

Package-based
- Commercial
- Open Source

SaaS **PaaS** **IaaS**

Cloud

IT Systems Main Acquisition Models

Custom-Built: software/system that is constructed (from "srcatch") according to an organization requirements/ specifications. Two main sourcing variations:

- In-house
- Outsourced/Offshored

Main Advantages	Main Issues
Flexibility — the software can be constucted/modified according to the specific business needs	Project risk and cost – the organization should assume the risks and costs of the project (including the quality of the result)
Control — the organization has limited dependency on 3rd parties (only partially on the outsourcing case)	Less-time-to-market (more time needed to develop the software)
Easier integration to other systems	Miss opportunity to obtain business know-how/adopt Best Practices (which are typically embedded in packaged software applications)
No license/maintenance fees (this may not be the case on the outsourcing scenario)	

IT Systems Main Acquisition Models

Package-based: Also known as "off-the-shelf" software. A ready-made software that implements a certain business functionality, for example, ERP and CRM. A packaged (ready made) software can be divided into two categories : **commercial** and **open source**.

- A commercial packaged software is built in order to be sold, i.e. is only available to an organisation that has agreed to pay a certain fee for it, typically
  (i) License fee (per number of package users or based on number of CPU's used on the servers that the package is installed)
  (ii) Annual Maintenance fee (for correction of errors and right to use the future versions of the package without licensing).

Its source code is typically not made available to the public.

- Open source is software whose source code is available to the public to use and change. Typically created in a collaborative way and improvements made are shared by the community. Common misconception: Open Source software is always free of charge – this may not be the case!. Different open source licenses exist, particularly for profit-organizations to have the right to use open source. Moreover, for organizations adopting open source a typical case is to pay annual support fees to a vendor supporting the software (e.g. Red Had, JBoss) – what is known as commercially supported open source.

IT Systems Main Acquisition Models

Package-based Commercial

Main Advantages	Main Issues
Faster time-to-market (project implementation)	Less flexibility (if the package needs to be modified (customized) to meet specific organizational needs – such customizations require extra fee and may lead to cost increase)
Opportunity to obtain business know-how/adopt Best Practices (which are typically embedded in packaged software applications)	Control — the organization has dependency on 3rd parties (the package vendor)
	License/maintenance/support fees
	Integration to other system may be more difficult (particularly if the package has not integration capabilities embedded)

IT Systems Main Acquisition Models

Package-based: Open Source (Vs Commercial)

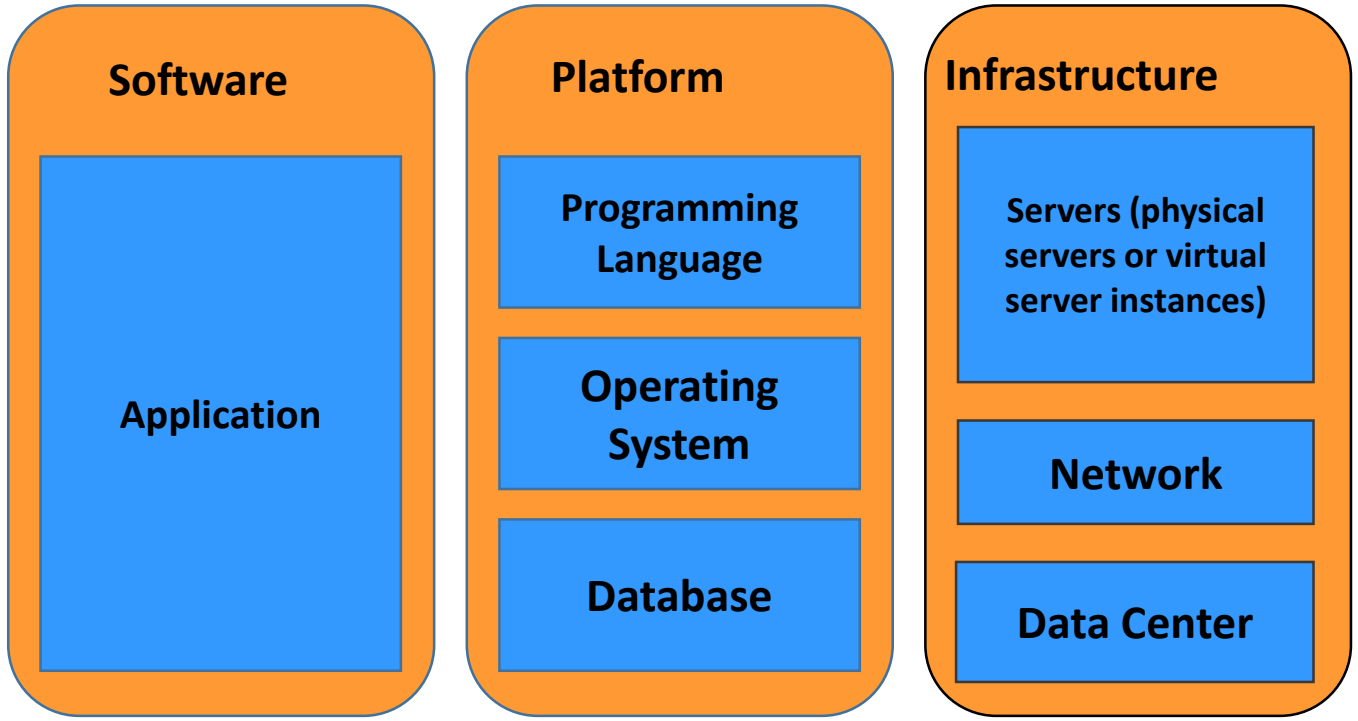
Main Advantages	Main Issues
Less cost (or at least less CAPEX (investment) cost - but this may not be the case for OPEX (annual) cost)	Support issues — potential difficulties to get support or feedback on the open source software.
Availability of source code	Maturity in the functionality provided
	Scalability (particularly for large installations)
	Legal/IP Issues

IT Systems Main Acquisition Models

Cloud: "*computer users (end users and organizations) conveniently rent access to fully featured applications, to software development and deployment environments, and to computing infrastructure assets such as network-accessible data storage and processing*."

3 main models:

- Software as a Service (**SaaS**)
- Platform as a Service (**PaaS**)
- Infrastructure as a Service (**IaaS**)

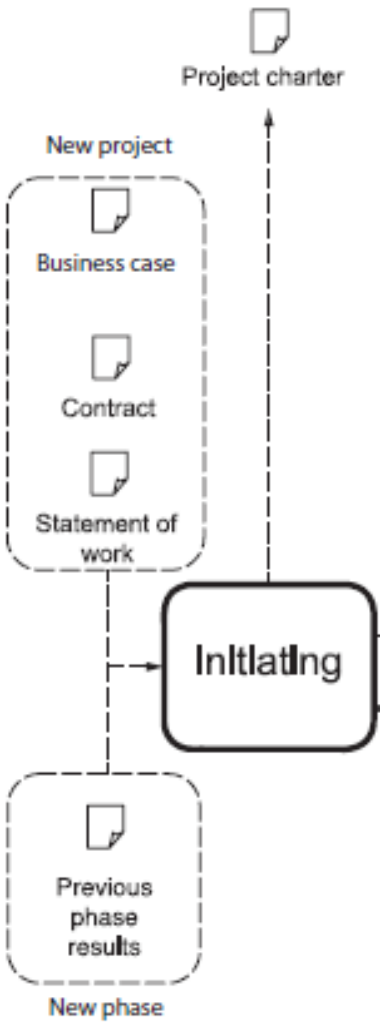


IT Systems Main Acquisition Models

Cloud

Main Advantages	Main Issues
More Flexible cost model (less CAPEX, emphasis on OPEX)	Vendor Lock-in, Loss of control
Operational responsibility to 3rd party (that typically has more expertise)	Security
	Integration
Less in-house sourcing requirements	
	Build expertise in cloud optimized solutions and cloud-related processes

Basics of Procurement and Contracts Management



Procurement: the process of acquiring products or services from a 3rd party.

Often done via a (formal) bid (tender) process.

Key Documents:

- Request For Information (RFI), - Request for Quote (RFQ)

- Request for Proposal (RFP): An invitation for suppliers of a product or service to submit a proposal (as part of a bidding process) in order to supply that product or service to the entity that issued the RFP.

Request for Proposal Template

I. Purpose of RFP

II. Organization's Background

III. Basic Requirements

IV. Hardware and Software Environment

V. Description of RFP Process

VI. Statement of Work and Schedule Information

VII. Possible Appendices

A. Current System Overview

B. System Requirements

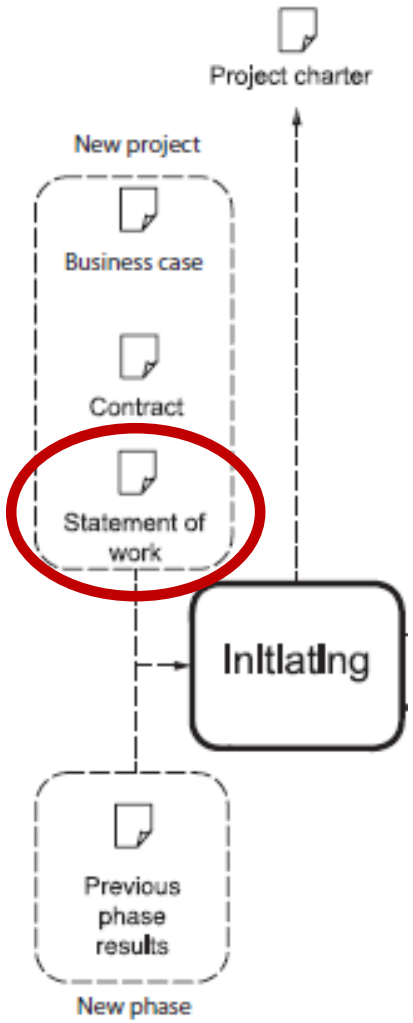
C. Volume and Size Data

D. Required Contents of Vendor's Response to RFP

E. Sample Contract

Image from Schwalbe's book .

Basics of Procurement and Contracts Management



Statement of Work: the description of work required for the procurement or as part of the contract.

A scope document with sufficient level of detail.

Clarity and non-ambiguity are critical factors.

Statement of Work (SOW)

I. **Scope of Work:** Describe the work to be done in detail. Specify the hardware and software involved and the exact nature of the work.

II. **Location of Work:** Describe where the work must be performed. Specify the location of hardware and software and where the people must perform the work.

III. **Period of Performance:** Specify when the work is expected to start and end, working hours, number of hours that can be billed per week, where the work must be performed, and related schedule information.

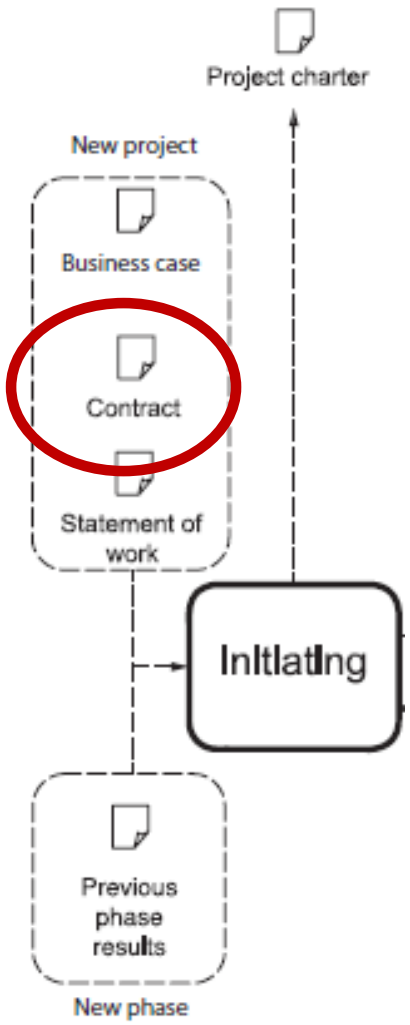
IV. **Deliverables Schedule:** List specific deliverables, describe them in detail, and specify when they are due.

V. **Applicable Standards:** Specify any company or industry-specific standards that are relevant to performing the work.

VI. **Acceptance Criteria:** Describe how the buyer organization will determine if the work is acceptable.

VII. **Special Requirements:** Specify any special requirements such as hardware or software certifications, minimum degree or experience level of personnel, travel requirements, and so on.

Basics of Procurement and Contracts Management



Main Types of Contracts:
Fixed Price (Lump Sum)
Fixed Total Cost Vs Fixed Unit Price
without incentive
with incentive
with price adjustment
Cost-reimbursable (aka Cost-Plus)
Cost plus Incentive Fee (CPIF)
Cost plus Fixed Fee (CPFF)
Cost Plus Award Fee (CPAF)
Cost Plus Percentage of Costs (CPPC)
Time and Material (T&M)